

SWIS: SELF-SUPERVISED REPRESENTATION LEARNING FOR WRITER INDEPENDENT OFFLINE SIGNATURE VERIFICATION

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Motivation

- A signature is constituted of several fundamental strokes, shifted by spatial coordinates and often varying in curvature.
- Each signature can be broken down to its constituent fundamental strokes which are decorrelated with each other.
- Fine-grained classification of decorrelated stroke information can help identify genuine and forged signatures.

Contributions

- A novel self-supervised approach is introduced for learning decorrelated stroke information signatures images for offline writer independent signature verification purpose.
- Results show that the proposed method is better or at par with the SimCLR, an InfoNCE loss based SSL framework and representations learning is also better in case of small datasets.

Pre-training Methodology

- Pre-processing step : We crop the smallest bounding box containing the signature.
- Reshape cropped signature to have dimensions 224 x 224 .
- Each signature is divided into patches and fed to the pretext encoder
- The output from the projector is used to calculate the loss and optimize the whole network.

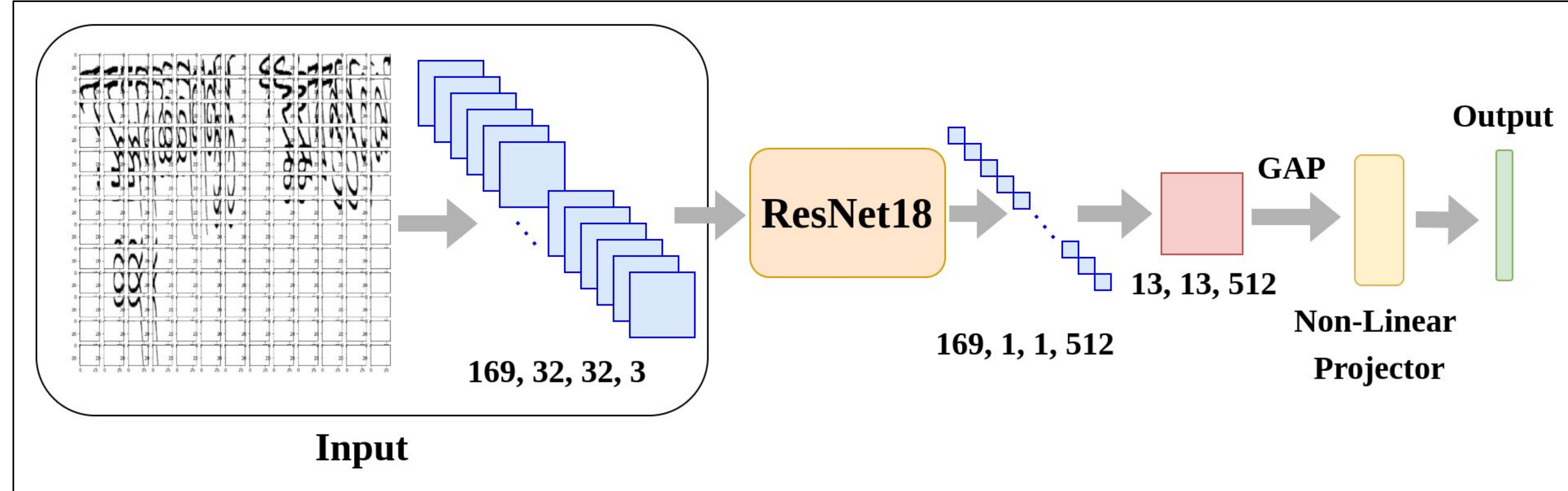
Downstream Task

- The downstream task is the writer independent offline signature verification task.
- To show the representation learning capability of the proposed framework, we use only SVM for the downstream task, train using the reference genuine signatures.
- If the classifier predicted a forged/genuine signature of a writer as a different/same writer, then we considered it a correct prediction.
- If the classifier predicted a forged/genuine signature of a writer as a same/different writer, then we considered it a wrong prediction.

Datasets

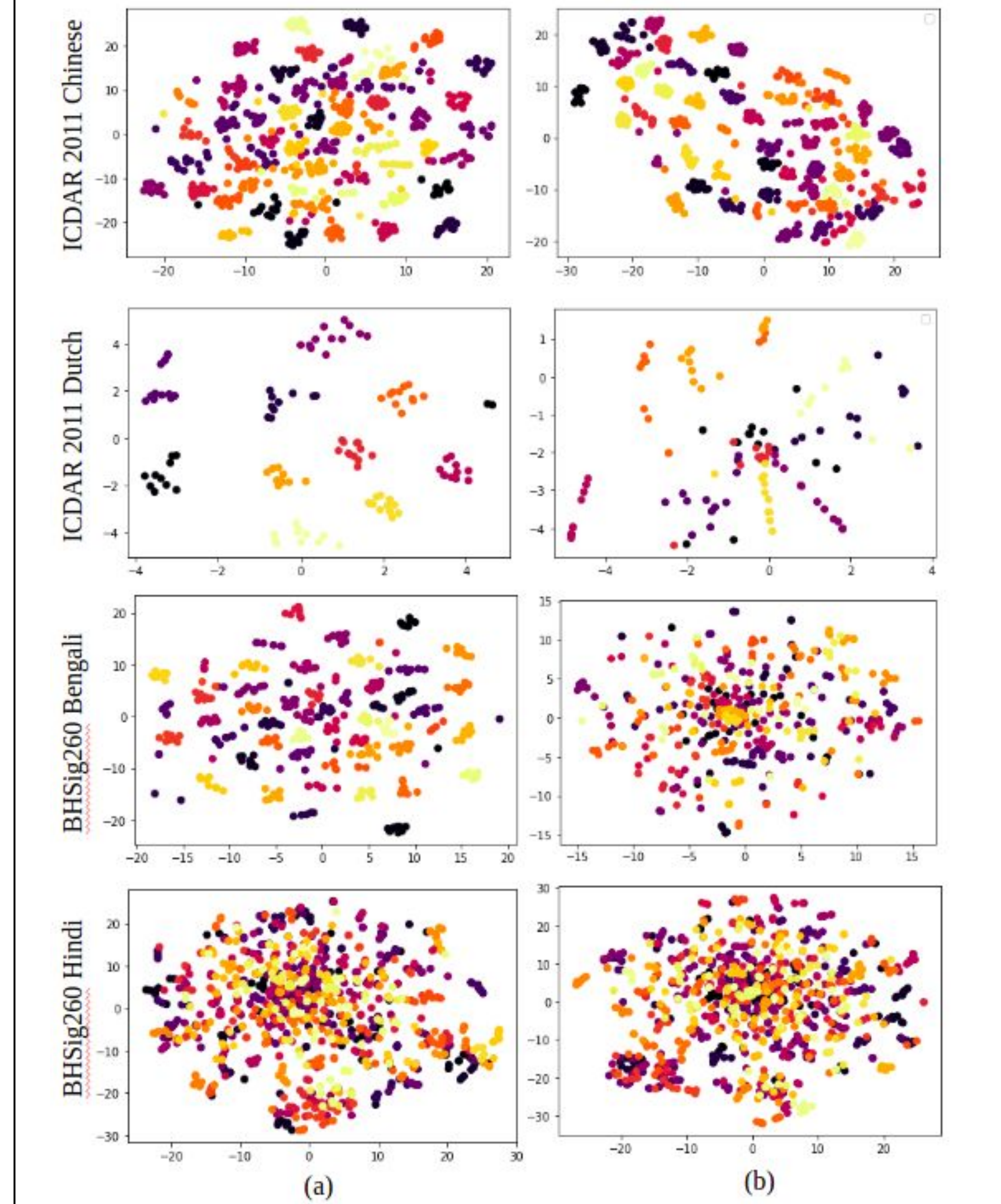
- BHSig260 Bengali, BHSig260 Hindi, ICDAR 2011 Dutch, ICDAR 2011 Chinese, CEDAR

Pre-training Model



Output Feature Representation

- The tSNE plots show the feature representation of our proposed framework (a) and SimCLR (b) [1].



References

1. T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in International conference on machine learning. PMLR, 2020, pp. 1597–1607.

Proposed Loss Function

- We calculate the cross-covariance matrix from the L2-normalized feature vectors obtained from the projector.
- The target for the cross covariance matrix is an Identity matrix
- The proposed loss function is given below

$$\mathcal{L}_C = \frac{1}{N} \sum_{i=1}^D \left(\sum_{j=1, j \neq i}^D \left(\sum_{k=1}^N z_k^i \cdot z_k^j \right)^2 + \left(\sum_{k=1}^N z_k^i \cdot z_k^i - 1 \right)^2 \right)$$

Comparison with SimCLR

- The comparison of the proposed framework with SimCLR on the downstream task is presented on the 4 datasets: BHSig260 Bengali and Hindi and ICDAR 2011 Dutch and Chinese is presented below

Method	ICDAR 2011 Dutch [18]			ICDAR 2011 Chinese [18]			BHSig260 Bengali [9]			BHSig260 Hindi [9]		
	Accuracy (%)	FAR	FRR	Accuracy (%)	FAR	FRR	Accuracy (%)	FAR	FRR	Accuracy (%)	FAR	FRR
SimCLR [4]	69.46	0.554	0.060	59.76	0.431	0.317	73.45	0.117	0.543	72.45	0.103	0.599
Proposed	77.62	0.316	0.133	64.68	0.278	0.583	72.04	0.367	0.116	72.43	0.104	0.598

Comparison with Supervised methods

- The comparison of the proposed framework with the supervised methods on the same datasets is presented below

Method	BHSig260 Bengali [9]			BHSig260 Hindi [9]		
	Accuracy (%)	FAR	FRR	Accuracy (%)	FAR	FRR
Pal et al. [9]	66.18	0.3382	0.3382	75.53	0.2447	0.2447
Dutta et al. [12]	84.90	0.1578	0.1443	85.90	0.1310	0.1509
Dey et al. [11]	86.11	0.1389	0.1389	84.64	0.1536	0.1536
Alaei et al. [13]	—	0.1618	0.3012	—	0.1618	0.3012
Proposed	72.04	0.367	0.116	72.43	0.104	0.598

Ablation Studies on CEDAR Dataset

Ablation type	Accuracy (%)	FAR(%)	FRR (%)
Best	83.8	11.8	18.7
Test Time AWGN $N(0, 0.01)$	76.84	32.42	17.0
Test Time Random Cropping	79.3	34.4	11.57

- Increasing Depth of Projector showed no change in performance
- Increasing pixel overlap from 0 to 8 pixels : Accuracy improved from 82.8% (0 pixels) to 83.8% (8 pixels).
- Removing Color Jitter Augmentation in Pre-training : Performance dropped by 1.7% accuracy.